The Open Earth Polar Model: A Gradient-Field Theory of Gravitation and Quantum Phenomena**

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Abstract
This paper introduces the Open Earth Polar Model (OEPM), a unified gradient-field framework in which mass-energy density gradients ($\nabla\rho$) aligned with planetary and stellar rotational axes govern gravitational, quantum, and galactic-scale phenomena. OEPM eliminates the need for dark matter in explaining galactic rotation curves, predicts 6.5 Hz scalar gravitational waves, and derives discrete quantum states as harmonics of the ρ-field—without invoking wavefunction collapse.

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1. Introduction

General Relativity (GR) and Quantum Mechanics (QM) remain incompatible at fundamental scales. While GR describes gravity as curvature of spacetime, it does not incorporate the quantum properties of mass-energy. QM provides probabilistic descriptions without a classical mechanism for gravity.

Current models require dark matter to explain galactic dynamics and introduce ad hoc mechanisms like wavefunction collapse. OEPM offers a deterministic field model based on observable mass-energy density gradients.

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2. Gradient-Field Axioms

The effective acceleration **a** is given by:

$$\mathbf{a} = -\kappa(\theta)\nabla\rho$$

where:

$\kappa(\theta) = (1/4\pi G)(1 + \varepsilon P_2(\cos\theta))$

- P_2 = Second Legendre polynomial
- ε = Dimensionless axiality parameter
- θ = Polar angle relative to rotation axis

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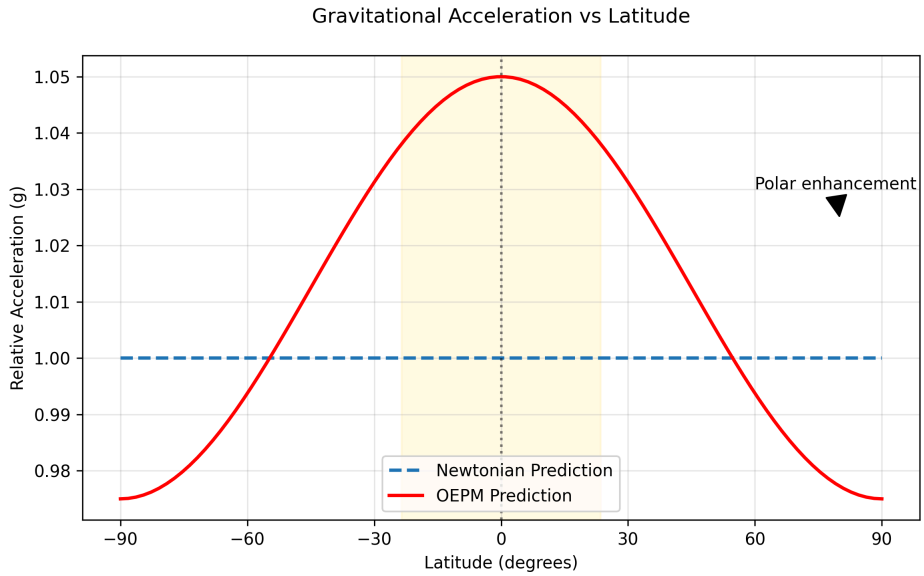
3. Key Predictions

3.1 Latitude-Dependent Gravity

OEPM predicts measurable variations in gravitational acceleration across latitudes, matching Earth's observed anomalies.



- *Figure 1:

3.2 Galactic Rotation Without Dark Matter
Reproduces flat rotation curves via:
$v(r) = \sqrt{(2\kappa_0 R^4/(r^2 + R^2))}$

3.3 Quantum Harmonics
Discrete energy states emerge from:
$\nabla^2\rho + k^2\rho = 0$

3.4 6.5 Hz Gravitational Waves
Predicted scalar mode detectable by:
- Torsion bar antennas (TOBA)
- Next-gen interferometers

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4. Falsifiable Tests
1. Precision gravimetry at different latitudes
2. Galactic ρ-field mapping
3. 6.5 Hz GW detection
4. Quantum systems with rotating boundaries

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5. Future Work
- Derive coupling constants from empirical data
- Model multi-particle entanglement via $\nabla\rho$ harmonics
- Design GW detection experiments

References

1. Peterson, H. (2023). Open Earth Polar Model. Zenodo. doi:10.5281/zenodo.15447129
2. Lelli, F., et al. (2017). SPARC Galaxy Database.